Etsy, 2024: nearest neighbours join keyword search

This weekend: Friday 08:00, the video store

Each search and each product is a list of numbers; the products nearest by inner product join the shortlist.

Result, A/B tests, relative

+2.63% *conversion rate of the whole site*

+5.58% *purchase rate of searches*

What it cost

On popular searches a plain version found 32% of the products bought; keyword search 63%.

What Etsy decided

A compressed index: under 4% of the nearest products missed, lookups within 20 ms.

Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024; preprint arXiv, June 2023.

Spotify, 2024: podcasts reveal audiobook tastes

This weekend: Friday 10:00, the matchmaker

Each listener and each audiobook is 128 numbers, learned from what people streamed, podcasts included.

Result, A/B test

+46.8% *listeners starting a new audiobook*

11.5M *monthly active users in the test*

What it cost

For users new to audiobooks its picks cover only 12.0% of the catalogue, against 19.3% for a simpler model.

What Spotify decided

In production for every eligible user; the narrower catalogue to be revisited.

De Nadai and others (Spotify), Companion Proceedings of the Web Conference (WWW) 2024.

Adyen, 2025: a better bandit approves more payments

This weekend: Friday 12:00, the bandits

Most payments can reach the bank in 2 to 15 ways. Epsilon-greedy picks the best bet, and now and then another.

Result, A/B test

+0.1% *approval rate, smarter exploring*

9M *more approvals a year, estimated*

What it cost

Exploring risks declines. Smarter exploring left fewer declines in the data: the next model did 0.2% worse.

What Adyen decided

Rules first: contracts and card network rules remove options before the bandit picks.

Vangara and Egg (Adyen), OARS workshop, KDD 2025. Egg (Adyen), Off-policy Evaluation for Payments at Adyen, arXiv, January 2025.

LinkedIn, 2024: bigger steps instead of longer training

This weekend: Friday 14:00, gradient descent

Models rank the feed, jobs and ads. The feed's has billions of parameters, trained on several GPUs.

Result, A/B tests, relative

+0.5% *member sessions in the feed*

+4.3% *click rate on ads*

What it cost

A training run took 70 hours; splitting the model across GPUs cut it to 20 hours. Many early runs diverged.

What LinkedIn decided

Longer training would slow experiments: triple the learning rate instead.

Borisyuk and others (LinkedIn), LiRank: Industrial Large Scale Ranking Models at LinkedIn, KDD 2024.

Which technique answers which question

The business question	*The technique*	*The cases*
Which products, or people, are nearest to this one?	Nearest neighbours	***Etsy, 08:00*** *also Airbnb, Snap*
Which unseen product will this client want?	Matrix factorization	***Spotify, 10:00*** *also Pinterest, Uber*
Which option to use, while still learning which works?	Bandits	***Adyen, 12:00*** *also Amazon, Dream11*
How to learn the numbers from billions of examples?	Gradient descent	***LinkedIn, 14:00*** *also Kuaishou*

Airbnb, arXiv 2026. Snap, SIGIR 2023. Pinterest, KDD 2022. Uber, engineering blog 2023. Amazon, AWS blog 2026. Dream11, arXiv 2023. Kuaishou, arXiv 2025.